

An Internet of Things System for Environmental Monitoring Based on ESP32 and Blynk

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Abstract— This study goes through basic principles of environmental monitoring in order to propose a simple real-time environmental monitoring based on the Internet of Things technology. The proposed solution utilizes inexpensive and widely available hardware and software components making it suitable for both personal and commercial use. The hardware of the sensor node is based on an ESP32 microcontroller equipped with sensors for environmental monitoring. The data is collected and integrated using Blynk's cloud-based web application as a backbone of the developed system. Blynk cloud platform provide features for storing, managing, and visualizing data received from monitoring device. The proposed system keeps track of air temperature, humidity, air pressure and dust-like particles concentration in the air. The system is characterized by low cost and low energy consumption. The sensor node has been installed and tested alongside a commercial system for ecological monitoring at the university building.

Index Terms— environmental monitoring, Internet of Things, Cloud platform, sensor nodes

I. INTRODUCTION

Environmental monitoring is about measuring aspects of the environment in a repetitive manner so that we can learn about its structure and functioning. Once we understand it better, we can apply that knowledge in multiple ways to manage it wisely [1]. Monitoring the environment parameters today is of big importance for nature and for our environments. Having knowledge over environment parameters can make us

adapt to nature, instead of nature adapting to us, and that can lead to unintended consequences and possible health risks for species living in it. In that domain, there are three basic disciplines of monitoring: soil, air, and water monitoring. The environment can be understood better by just knowing how some parameters change after human interactions with the environment.

Ecological and environmental monitoring is a slow and repetitive process, and it takes a long time to have some insightful information and knowledge. The development of information technology has allowed us to collect environmental data as the processes unfold and repeat the measurement process periodically for long periods [1,2]. The main problem is that the data is collected in a variety of formats and there are no precisely defined standards that can help in keeping a unified knowledge base about environmental and ecological data, even at the national level. It is not uncommon that we are often unaware of the existing databases in our region or in different parts of the world [3].

Information technology components and systems are becoming increasingly small and more affordable, more powerful, and therefore frequently used. People no longer only encounter information technology at common points in their lives, but as information and communication infrastructures, which are present in all areas of our everyday lives [4]. As the Internet of Things (IoT) concept is overtaking the tech world, it opened various possibilities for integrating information technologies into many business domains and processes, as well as it helped in creating new technology solutions with a lot of opportunities. IoT is one of the main development and innovation tools industry disruptive concept such as Industry 4.0 [5].

The main feature of the IoT is communication between the devices connected to the Internet, with no human interaction needed, while they process data and exchange it with each other [6]. The data can also be communicated to remote locations for further aggregation and analysis to create new knowledge needed to improve the decision-making process. As such, IoT technology provides various possibilities

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when it comes to eliminating redundant work and processes to make our lives easier and our environment a better place. IoT enables a more informative approach to everyday life encounters, such as finding parking spot, fitness, weather conditions, breathing cleaner air, producing healthier food, etc [7].

In this paper, we are proposing an IoT system for environmental monitoring based on a low-cost ESP32 microcontroller and an IoT cloud platform. Main aspects of the proposed solution are an ease of implementation, the use of inexpensive components, and extremely low power consumption. As such, it can be utilized for both personal and commercial use and the solution can easily be upgraded and scaled. The cloud platform is used for storing and analyzing data received from IoT sensor nodes and it provides means for integration into mobile apps and creation of data and knowledge dashboards for visualization, which makes the system easy to use and keep track on measurements.

II. MATERIALS AND METHODS

A. Sensor Node Microcontroller (ESP32 MCU)

The sensor node is designed around the ESP32, a microcontroller board developed by Espressif - a firm dedicated to development of AIoT (Artificial Intelligence of Things) chips, with built-in Wi-Fi and Bluetooth modules [8]. ESP32 is one of the most powerful chips made by Espressif available on market. In relation to chips before it, ESP32 is considered as improved version on many aspects. Besides having WiFi support, it has a Bluetooth communication module making it able to communicate with Bluetooth compatible devices. It has the same processor chip as its predecessor ESP8266, but with two additional cores, and higher speed. This version has more data memory, more general-purpose input/output (GPIO) pins. Another advancement worth mentioning is that ESP32 has a Controller Area Network (CAN) bus connectivity, which previous model ESP8266 does not have [9]. The functional block diagram of ESP32 is given in Fig. 1 [10].

B. IoT data platform (Blynk Cloud)

As for the IoT data platform, we opted for Blynk. It is a cloud based system designed for concept of IoT, providing a platform for easy data managing and storing data in the cloud, making it a part of technologies that support IoT. This means that most of IoT programmable devices can be connected to it, and therefore use the benefits and capabilities this platform

offers. Blynk contains app builder for mobile phone and its libraries are made available free of charge. In addition, it provides a legacy open-source server app, which is off-limits and can be installed locally, and therefore, hundreds of devices could be easily managed at no cost [11]. Free cloud service is managed through a system of points which are spent through creating dashboards on mobile, as well as in web application, where data received is visualized. In this study, Blynk is used as a middleware through which all the communication and data sharing is made possible [11].

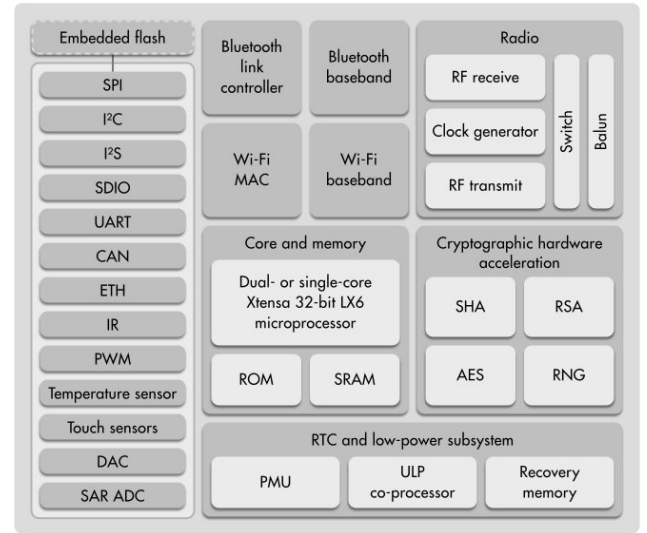


Fig. 1. ESP32 Function Block Diagram [10]

C. Sensors and Monitoring Parameters

The ESP32 based sensor node is equipped with sensors for monitoring air temperature, air humidity, air/water temperature, atmospheric pressure, and dust particles. The selection of sensors and their characteristics are provided in Table 1 [12-15]. The table shows the measuring ranges and accuracy of the sensors used. DHT22 sensor's functions were used only for measuring air humidity parameters in this study, as there is a DS18B20 temperature sensor, and is waterproof, which makes it a good choice for outside environmental monitoring. DHT22 has a good measuring range, as well as precision (same as DS18B20), but the choice of DS18B20 was only due to its waterproof characteristics. Atmospheric pressure sensor, BMP180 has good precision which made it a good choice for measuring air pressure parameters. And the last, dust concentration sensor, GP2Y1014AU0F, which can diversify smoke from dust particles in the air. It is highly accurate when it comes to recognizing particles of at least the size of 0.5 μ m. This sensor considers anything that is not usually found in the air

and returns the particles concentration level in the air in $\mu\text{g}/\text{m}^3$. This sensor is also qualified as an air quality sensor, as it shows zero values when the air is clear. It finds use in many air purifier systems. In this study it was used solely to represent the state of the particles concentration in the air.

TABLE I
SENSOR CHARACTERISTICS

Sensor	Measuring range	Accuracy
Temperature - DHT22	-40 to 80°C	$\pm 0.5^\circ\text{C}$
Humidity - DHT 22	0 to 100%	± 2 to 5%
Temperature - DS18B20	-55 to 124°C	$\pm 0.5^\circ\text{C}$
Atmospheric pressure - BMP180	300 to 1100hPa	$\pm 0.12\text{hPa}$
Dust particles - GP2Y1014AU0F	0 to 1500 $\mu\text{g}/\text{m}^3$	$\pm 15\%$

The sensor node is collecting measurements periodically, transforms them into usable information, and sends them to the cloud platform, where this information is stored and visualized in a user-friendly fashion. It is important to note that due to the flexibility of the selected microcontroller technology, there are many opportunities to expand this sensor node and equip it with additional or different sensors for monitoring all kinds of different environment-related parameters. When it comes to integration of the sensors with the ESP32 board there were no limitations, except the number of ports, as the ESP32 board supports digital and analog inputs (ADC) [8,10].

The overall advantages in using selected sensors is that they are easy to connect and power up, and they come factory calibrated. The precision is good enough for the use they are intended to, meaning for the creation of inexpensive and effective monitoring systems. The sensors are easy to connect and easy to program as there are pre-built software libraries for each of them. The system is easy to maintain and extremely low cost.

III. IMPLEMENTATION AND DISCUSSION

In this section, the process of integrating the components is described, with an in-depth look at the created system. At last, the selected technologies are compared to their alternatives, and further discussion on the implementation approach is given.

The implementation approach is illustrated with a block diagram in Fig. 2. The diagram shows the principle of this system's overall function. The central control unit of an IoT sensor node, in this case, ESP32 MCU, periodically reads sensors for change in state. When the MCU gets the input from the sensor, it decodes it into valuable data, and through

communication with Blynk API libraries, that handle requests, sends the data to the IoT data platform (Blynk cloud). The mobile cloud app is connected to Blynk cloud and is picking up changes in the cloud that are the result of uploading new data from the IoT sensor node. The mobile app is also provided by the Blynk platform solution.

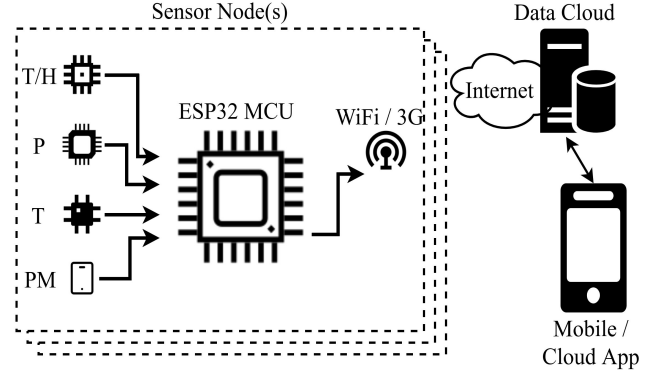


Fig. 2. Implementation approach: components and data flow view

Articulating the data from the sensors is achieved using the official software libraries, so there is not much coding needed to activate sensors and obtain the measurement. ESP32 microcontroller is software compatible with Arduino, and Arduino integrated development environment (IDE) was used for code development. As stated, the programming process was not that demanding, since all the sensors have their official libraries available for use within Arduino IDE. Sensor measurement readings are easily transformed into appropriate data. As for the Blynk, it provides data integration libraries that give full support for sending data to the cloud directly from the microcontroller. The Blynk libraries handle all needed protocols for implementing the communication with Blynk cloud. For this study, WiFi connection was used for connecting to Internet, and to make it possible for ESP32 to communicate with Blynk data integration platform. Optionally, the system can be configured to use the mobile network.

ESP32 MCU is provided with enough I/O pins to handle all the digital or analog inputs from the sensors used, but the number of sensors and use of WiFi especially, requires a stable external power source which can handle constant electrical needs. Powering the sensors through ESP32 would be too demanding and it could result in sensors not working or in their non-accurate inputs due to under-voltage or other power supply issues. For this sensor node, a commercial power supply of 5V was used (Meanwell) and ESP32 was powered through it, as well. Another

good alternative would be the use of a solar panel and battery, but as this system was installed near an electric outlet, the power supply was used.

Connecting sensors to the board was a straightforward process. DHT22 VCC pin demands at least 3.3V and it is connected to the breadboard that is supplied with electric power from the power supply. The data pin of DHT22 was connected to the D4 data pin on ESP32. The ground pin is also connected to the breadboard with the power supply's ground. As for DS18B20, the VCC pin is also in need of at least 3.3V of electrical power and is connected to the breadboard. DQ pin is connected to the D16 pin of ESP32. DQ pin should be pulled high, so resistor of 5k Ω is set between power source and data (DQ) pin. The BMP180 air pressure sensor is somewhat more complicated than the last two because it requires 2 pins for input communication, and 5V of power. Usually, the VIN pin of BMP180 is connected to the VIN pin in ESP32 when the USB power-on is used, but for this case, it is connected to the breadboard. SDA and SCL pins are connected to D21 and D22 pins on ESP32 respectively. The ground is connected to the breadboard. Finally, when interfacing GP2Y1014AU0F dust sensor and ESP32 only needed pins are VP, for analog input, and one pin to initiate the led inside it, the D12 pin was chosen on ESP32, and the VCC pin of the sensor was connected to the board. The analog keeps track of light refraction which is then processed by the board through an analog input. As mentioned before, the programming of the ESP32 MCU to access the sensors was an easy process thanks to the software libraries available for all the selected sensors:

- Adafruit DHT sensor library (for DHT22),
- OneWire and DallasTemperature libraries used for DS18B20,
- BMP180MI library was used for BMP180,
- Sharp GP2Y Dust Sensor library was used for interfacing the dust sensor and ESP32.

Having Blynk interact with ESP32 was done through available Blynk libraries and APIs [11]. The whole process of sending and requesting data inputs is done through them. All it was needed is to acquire authentication tokens, which recognize the preconfigured device through the Blynk platform, and therefore allow their interaction, i.e., sending the sensor data from ESP32 MCU to the cloud. The data is visualized using Blynk mobile dashboard as illustrated in Fig. 3. The user can monitor sensor measurements, parameter changes and trends, using the mobile app provided by the Blynk platform.

There are many alternatives to technologies used in this study, and they are in a way, alternatives to some other technologies. Like ESP32 is to Arduino. Arduino architecture and its microcontroller units are most popular in terms of building and prototyping IoT projects, and it goes along with how many versions of Arduino-based microcontrollers are there. Arduino microcontrollers with built-in WiFi, use built-in ESP8266 technology. ESP32 is typically compared with some beginner versions of Arduino such as Uno and Zero, but its specifications almost make it more powerful than most commercially available Arduino boards. In Table 2, a comparison between the ESP32 board, used in the proposed system, and Arduino Mega 2560 is considered one of the top MCU boards of the Arduino family [8,16].

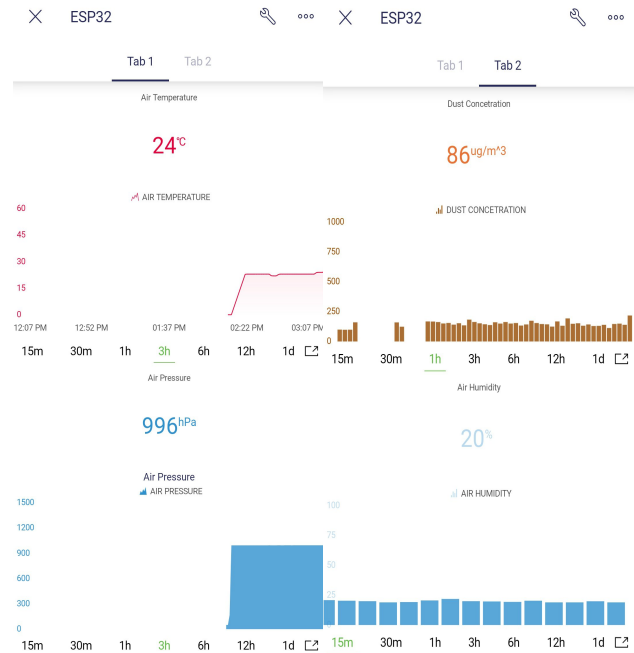


Fig. 3. Mobile dashboard for visualization of sensor data

TABLE II
COMPARISON BETWEEN ESP32 AND ARDUINO MEGA

Characteristics	ESP32	Arduino Mega 2560
Operating voltage	2.6-3.3V	5V
Architecture (bits)	32bit	8bit
Input voltage	3-5V	7-12V
General purpose I/O pins	36	70
Flash memory	16MB	256KB
SRAM	512KB	8KB
Clock Speed	160-240MHz	16MHz
WiFi	Yes	No
Bluetooth	Yes	No

As for the IoT platform, there are various alternatives to Blynk available [17,18]. There is also an option for in-house implementations, but this may not be practical

for quick and inexpensive solutions [2,19]. Lately, one interesting alternative is the Arduino Cloud platform [20]. Its concept is similar to Blynk, but it does not offer the same flexibility. It has a mobile cloud remote app but does not support creating dashboards from the mobile app. On one hand, it provides a cloud programming tool (sketch editor), which enables programming the device through the cloud without adding any additional libraries to the code. On the other hand, it does not provide a mobile app builder or server app, and it has more restrictions in designing dashboard interfaces than Blynk. Blynk has a holistic approach to devices managing and offers easy ways to connect and visualize data. That was the main reason why it was selected as a developing tool for this study.

The proposed system can be easily adapted by adding or replacing selected sensors with other types of sensors. Also, the system can be scaled by adding additional sensor nodes, which makes it suitable for a wide variety of use cases. Besides indoor or outdoor ecological and environmental monitoring, such a system could easily be adapted for applications in precision agriculture, smart home, smart city, and other domains [21].

IV. CONCLUSION

This paper describes an implementation of a low-cost low power IoT system for environmental monitoring using ESP32 microcontroller and Blynk cloud platform. The paper discusses the selection of components, approach to the system implementation, and initial experiences with the system. The proposed system can be used for ecological and environmental monitoring, and it can easily be extended using different sensors. The future work will include implementation with power supply based on the use of solar panel and batteries, as well as experimentation with various sensors that can support other use cases, especially in domains of precision agriculture and smart cities.

Future research will include a long-term evaluation of the system and a comparison of the sensor data with the widely used commercial systems for ecological monitoring. In addition, we will explore the customization of the system for its use in precision agriculture. The customization will consider power supply based on solar panel and battery, as well as experimenting with a variety of sensors.

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